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SE3070 — Case Studies in Software Engineering

Assignment 01: Interaction and Functional Design of a Software Solution

Smart Wildlife Conservation and Anti-Poaching Monitoring System

Case Study 1 — Wildlife Conservation System

A digital platform for the Department of Wildlife Conservation of Sri Lanka

Group ID: CSSE_NU_WE_01

Name	Registration Number	Individual Use Case Scenario
Joel Nithushan A.T	IT23556652	UC01-S01 – Conduct Assigned Ranger Patrol
Thushalini U	IT23794870	UC02-S01 – Report Field Incident
Vaishnavi L	IT23717336	UC03-S01 – Monitor Tracked Wildlife and Manage Risk Alerts
Kanistan T	IT23748644	UC04-S01 – Manage Human-Wildlife Conflict Reports

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1 Introduction

This report presents the group and individual functional-design deliverables for the Smart Wildlife Conservation and Anti-Poaching Monitoring System. The group selected four distinct major business use cases, and each member developed one detailed scenario, sequence diagram, storyboard, low-fidelity wireframe and high-fidelity wireframe for their assigned use case.

1.2 Team and Use Case Allocation

Each member worked on a different selected use case. The allocation is shown below.

Member	Student ID	System use-case area (Part A)	Individual use case scenario (Part B)
Joel Nithushan A.T	IT23556652	UC01 – Manage Ranger Patrol	UC01-S01 – Conduct Assigned Ranger Patrol
Thushalini U	IT23794870	UC02 – Report and Manage Field Incident	UC02-S01 – Report Field Incident
Vaishnavi L	IT23717336	UC03 – Monitor Tracked Wildlife and Manage Risk Alerts	UC03-S01 – Monitor Tracked Wildlife and Manage Risk Alerts
Kanistan T	IT23748644	UC04 – Manage Human-Wildlife Conflict Reports	UC04-S01 – Manage Human-Wildlife Conflict Reports

Part A – Group Deliverables

2 High-Level Use Case Diagram

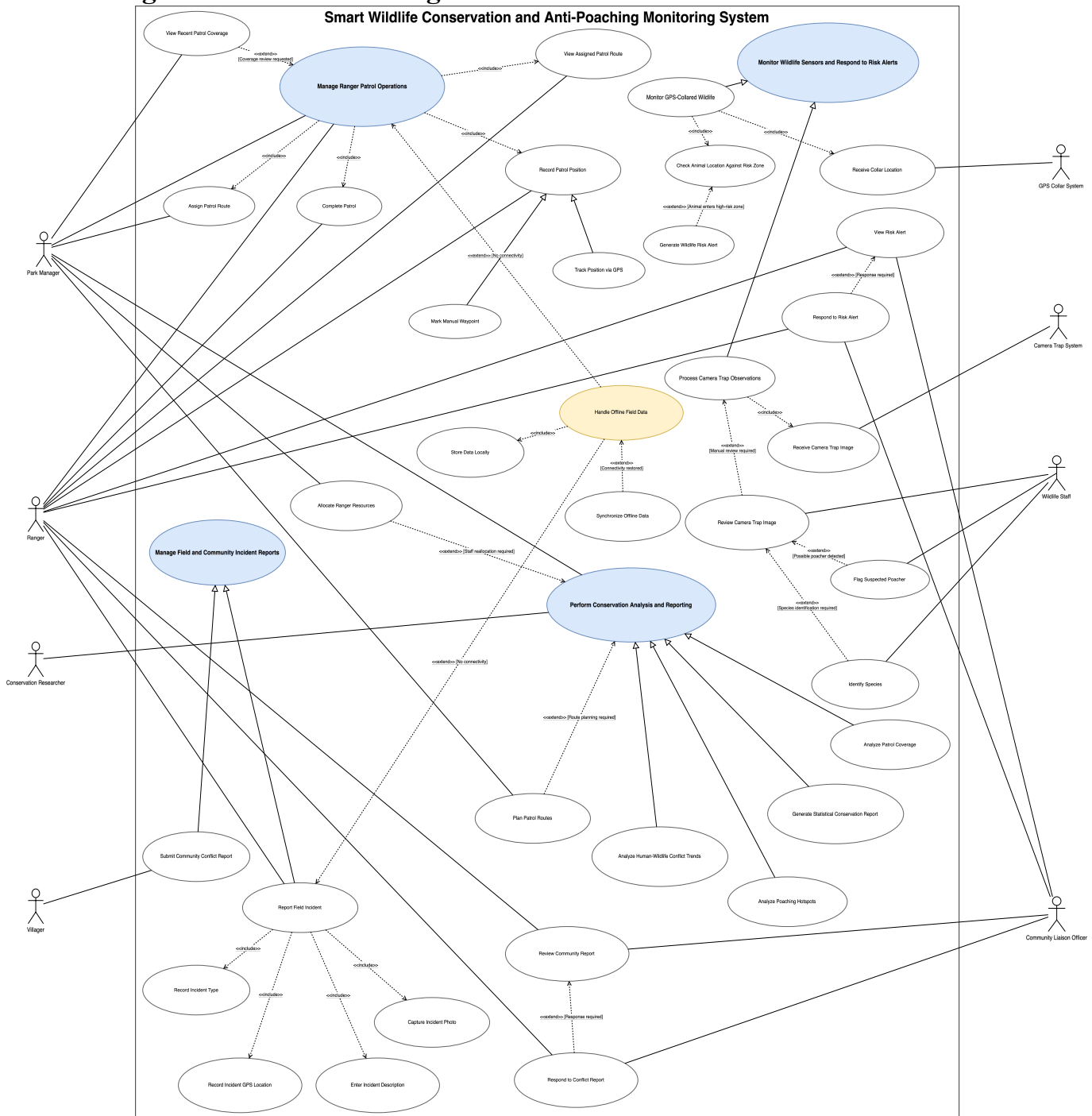


Figure 1 – High-level use case diagram for the four selected business use cases

Part B – Individual Deliverables

4 Joel Nithushan A.T (IT23556652) – UC01-S01 Conduct Assigned Ranger Patrol

4.1 Use Case Scenario – UC01-S01 Conduct Assigned Ranger Patrol

Use Case ID	UC01-S01
Use Case Name	Conduct Assigned Ranger Patrol
Parent Use Case	UC01 – Manage Ranger Patrol
Goal / Description	This scenario describes how a Ranger conducts a patrol that has already been assigned by a Park Manager. The Ranger views the assigned patrol route, starts the patrol, and follows the route while the System records the Ranger's position, either automatically through GPS or as manually marked waypoints. If network coverage is lost in the field, the System stores position data locally and synchronises it once connectivity is restored. When the route has been covered, the Ranger completes the patrol and the System records the completion and updates the patrol coverage for that route.
Primary Actor	Ranger
Secondary Actor(s)	None
Trigger	The Ranger is ready to begin an assigned patrol and opens it on the field application.
Preconditions	P1. A valid Patrol has been assigned to the Ranger by a Park Manager (Assign Patrol Route has already occurred). P2. The Patrol is associated with an active PatrolRoute. P3. The Ranger has access to the assigned patrol information on the field application. P4. The Patrol status permits it to be started (it has not already been started or completed). P5. The Ranger's device has GPS/location capability enabled, although individual GPS readings may occasionally be unavailable.
Postconditions – Success	S1. The Patrol status is recorded as completed. S2. The Patrol completion time is stored.

	S3. All captured PatrolPosition records, both GPS and manual, are associated with the Patrol. S4. Any PatrolPosition records stored locally while offline have been synchronised, or remain pending with no data lost, ready to synchronise when connectivity is next available. S5. Patrol coverage for the PatrolRoute has been calculated from the recorded positions and is available for Park Manager review.
Postconditions – Minimal Guarantee	F1. Every PatrolPosition successfully captured during the patrol is preserved and is not lost, including positions recorded while offline. F2. The Patrol remains in a valid, consistent status at all times.

Main success scenario

Assumes normal network connectivity throughout. Alternative and exception flows state the step from which they branch.

Step	Actor	Action
1	Ranger	Opens the assigned patrol on the field application.
2	System	Displays the assigned PatrolRoute (name, description, planned route on the map) and the current Patrol details, including its assigned time and status.
3	Ranger	Reviews the route and selects the option to start the patrol.
4	System	Starts the Patrol, records the start time, changes the Patrol status to in progress, and begins position tracking.
5	System	While the Ranger follows the route, periodically obtains the Ranger's current GPS position.
6	System	Records each valid GPS position as a PatrolPosition with capture mode GPS, associates it with the active Patrol, and shows the covered track on the map.
7	Ranger	Continues along the assigned PatrolRoute; steps 5–6 repeat throughout the patrol until the route is covered.
8	Ranger	Reaches the end of the route and requests to complete the patrol.
9	System	Records the completion time and changes the Patrol status to completed.

Step	Actor	Action
10	System	Calculates the patrol coverage from the PatrolPositions recorded during the Patrol and updates the coverage for the PatrolRoute.
11	System	Confirms successful completion to the Ranger, showing the patrol duration, number of positions recorded, and coverage achieved.

Alternative flows

A1 – Manual Waypoint <i>Branches from step 5–7 of the main flow, when the Ranger decides a location should be explicitly recorded.</i>	A1.1 The Ranger notices a location that should be explicitly marked on the patrol record, for example a boundary point, a water source, or an area of interest. A1.2 The Ranger selects the option to mark a waypoint. A1.3 The System captures the current coordinates and the time of capture. A1.4 The System creates a PatrolPosition with capture mode MANUAL. A1.5 The System associates the PatrolPosition with the active Patrol and shows the waypoint on the covered track. A1.6 The flow returns to step 7 of the main flow.
A2 – No Network Connectivity <i>Branches from any point between steps 4 and 8 of the main flow. Consistent with Handle Offline Field Data «extend» Manage Ranger Patrol Operations [No connectivity], which includes Store Data Locally.</i>	A2.1 The System detects that network connectivity is unavailable. A2.2 The System shows an offline indicator to the Ranger and continues operating. A2.3 The Ranger continues the patrol normally. A2.4 The System continues capturing GPS positions and manual waypoints as PatrolPositions. A2.5 The System stores each captured PatrolPosition locally on the device. A2.6 The System marks each locally stored PatrolPosition as pending synchronisation and shows the pending count to the Ranger. A2.7 The Ranger continues the patrol with no loss of collected field data. A2.8 The flow returns to the main flow at the point where connectivity was lost; if connectivity is restored during the patrol, Alternative Flow A3 applies.
A3 – Connectivity Restored <i>Branches from Alternative Flow A2 when the network</i>	A3.1 Network connectivity becomes available again. A3.2 The System identifies the locally stored PatrolPositions that are pending synchronisation.

<i>becomes available again. Consistent with Synchronize Offline Data «extend» Handle Offline Field Data [Connectivity restored].</i>	A3.3 The System synchronises the pending PatrolPositions with the central patrol record. A3.4 The System marks each successfully synchronised PatrolPosition as synchronised and reduces the pending count accordingly. A3.5 The Ranger continues the patrol; no field data needs to be re-entered. A3.6 The flow returns to the main flow. If connectivity is not restored before the patrol ends, the pending positions are synchronised the next time connectivity is available (see Postcondition S4).
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Exception flows

E1 – GPS Position Cannot Be Obtained <i>Branches from step 5 of the main flow.</i>	E1.1 The System cannot obtain a valid GPS position (for example under dense canopy or inside a ravine). E1.2 The System informs the Ranger that the automatic position could not be recorded and keeps the previously recorded track intact. E1.3 The Ranger may move on and allow the System to retry automatically, or manually mark a waypoint at the current location (Alternative Flow A1). E1.4 If a manual waypoint is successfully captured, or a subsequent GPS reading succeeds, the flow returns to step 7 of the main flow. E1.5 The Patrol continues in its in-progress status; a failed reading does not end the patrol.
E2 – Patrol Completion Cannot Be Recorded <i>Branches from steps 8–9 of the main flow.</i>	E2.1 The Ranger requests to complete the patrol. E2.2 The System is unable to record the completion action at that moment. E2.3 The System informs the Ranger that completion has not yet been recorded. E2.4 The System preserves all PatrolPositions recorded during the patrol. E2.5 The Patrol remains in its previous valid status (in progress) until the completion can be recorded successfully. E2.6 The Ranger may retry the completion request; on success, the flow resumes at step 9 of the main flow.

4.2 Sequence Diagram – UC01-S01

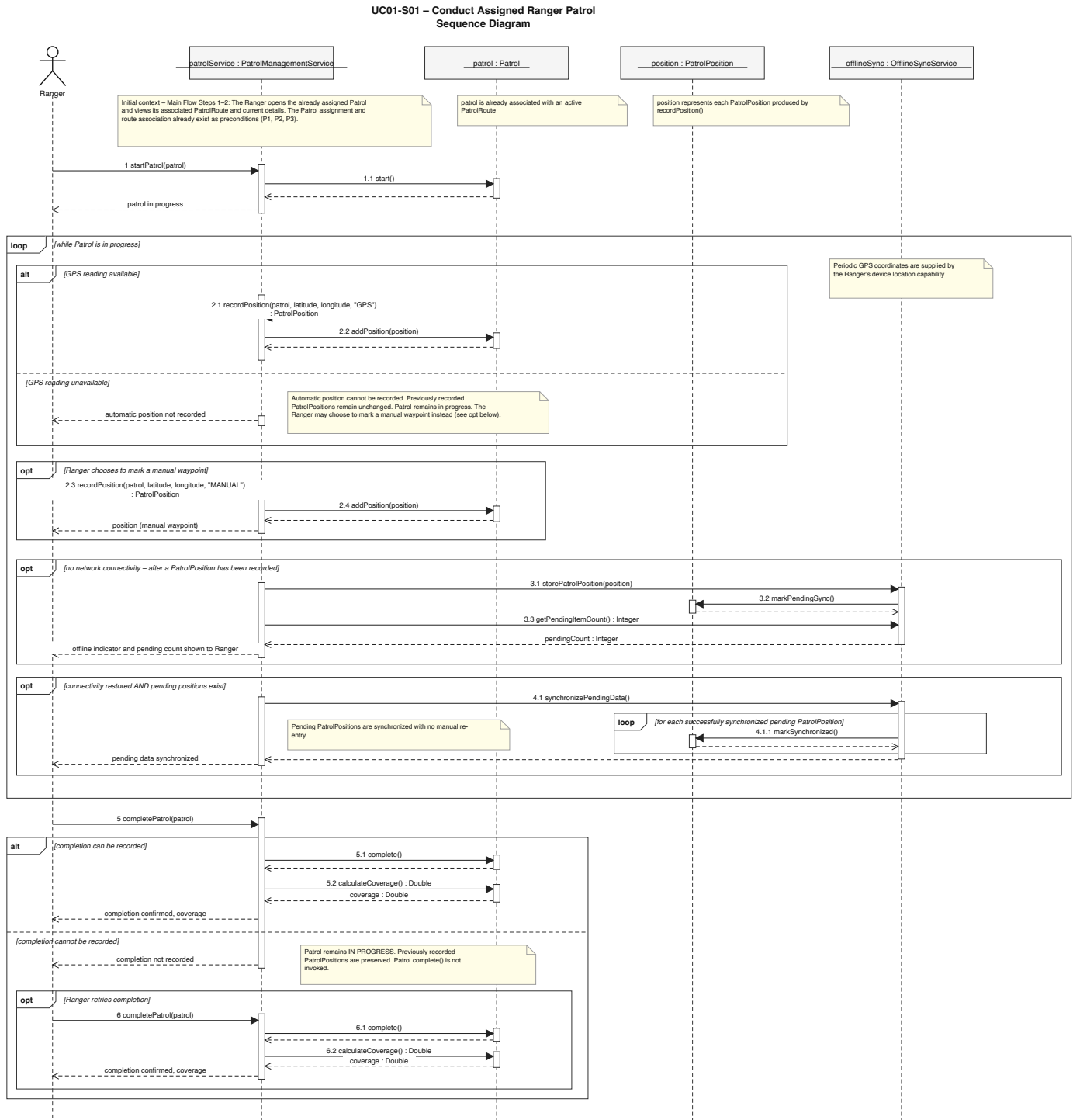


Figure 3 – Sequence diagram for UC01-S01 Conduct Assigned Ranger Patrol

4.3 Hand-Drawn Storyboard – UC01-S01

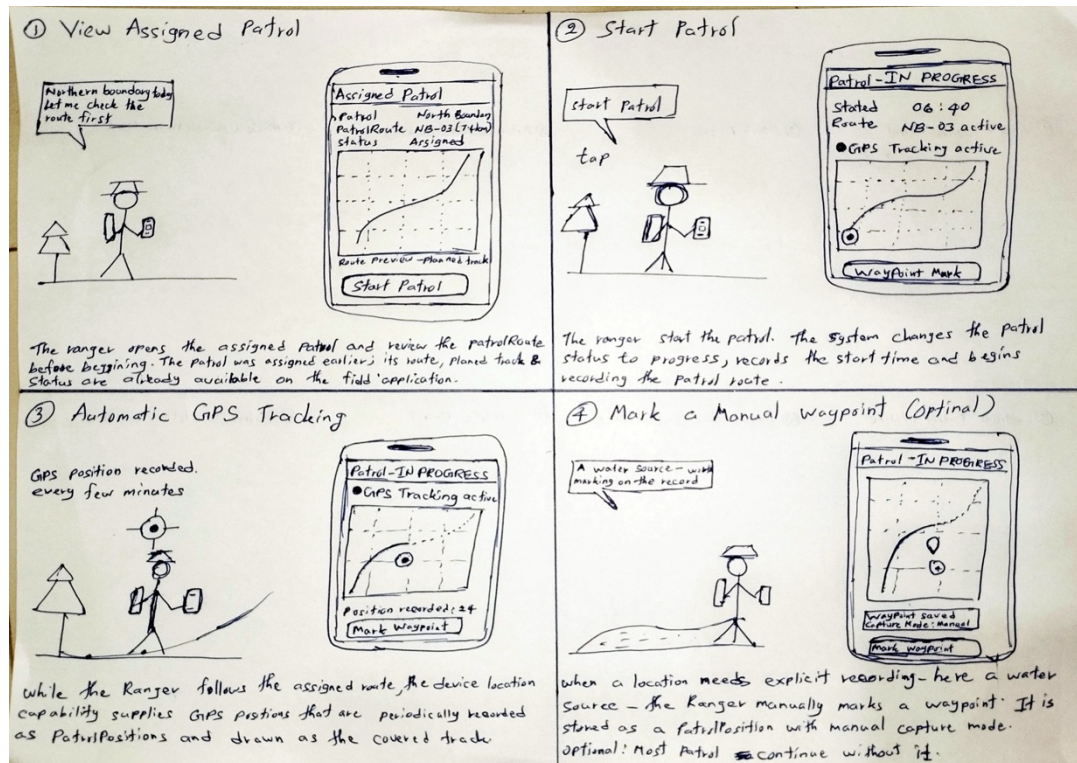


Figure 4(a) – Hand-drawn storyboard for UC01-S01 (panels 1–4)

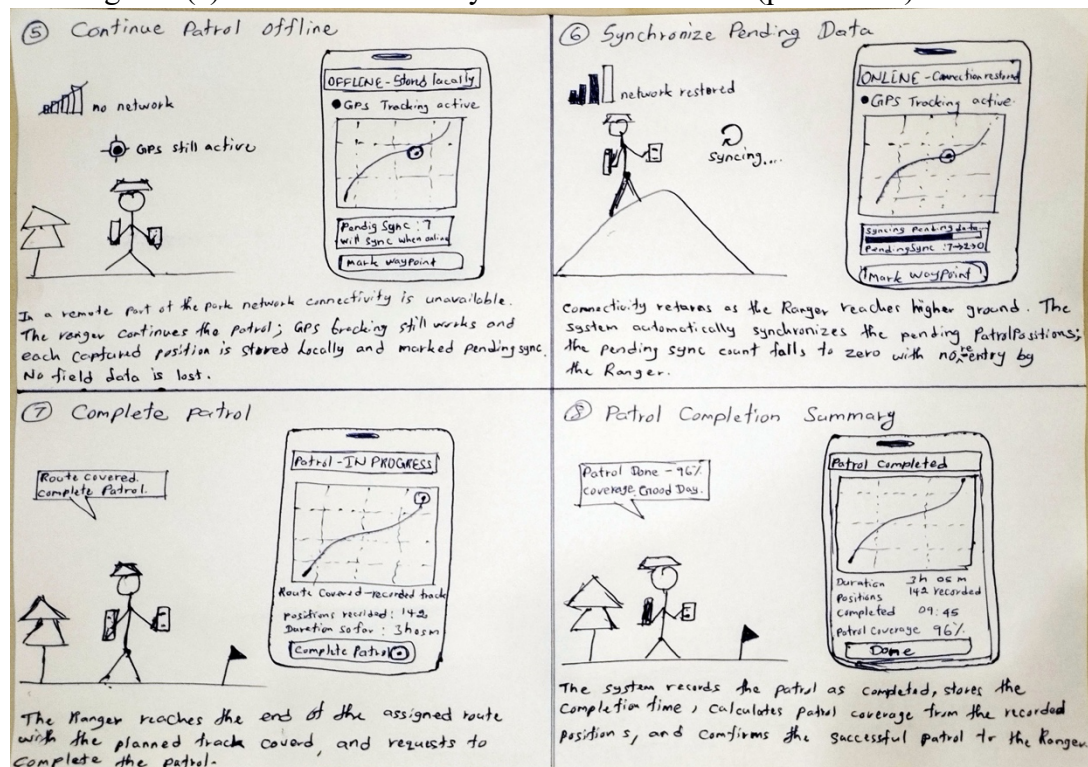


Figure 4(b) – Hand-drawn storyboard for UC01-S01 (panels 5–8)

4.4 Low-Fidelity Wireframe – UC01-S01

UC01-S01 – Conduct Assigned Ranger Patrol

Low-Fidelity Wireframe – skeleton (screen rectangles 390 × 844, no device frame)

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Skeleton low-fidelity: layout, hierarchy, component placement and navigation only – grey bars stand for text, crossed boxes for images / maps, filled rectangle = primary action, outlined = secondary, dashed = conditional. Screens are plain rectangles without a device frame. Content, labels and colour are defined in the high-fidelity wireframe of the same scenario.

Figure 5 – Low-fidelity wireframe for UC01-S01

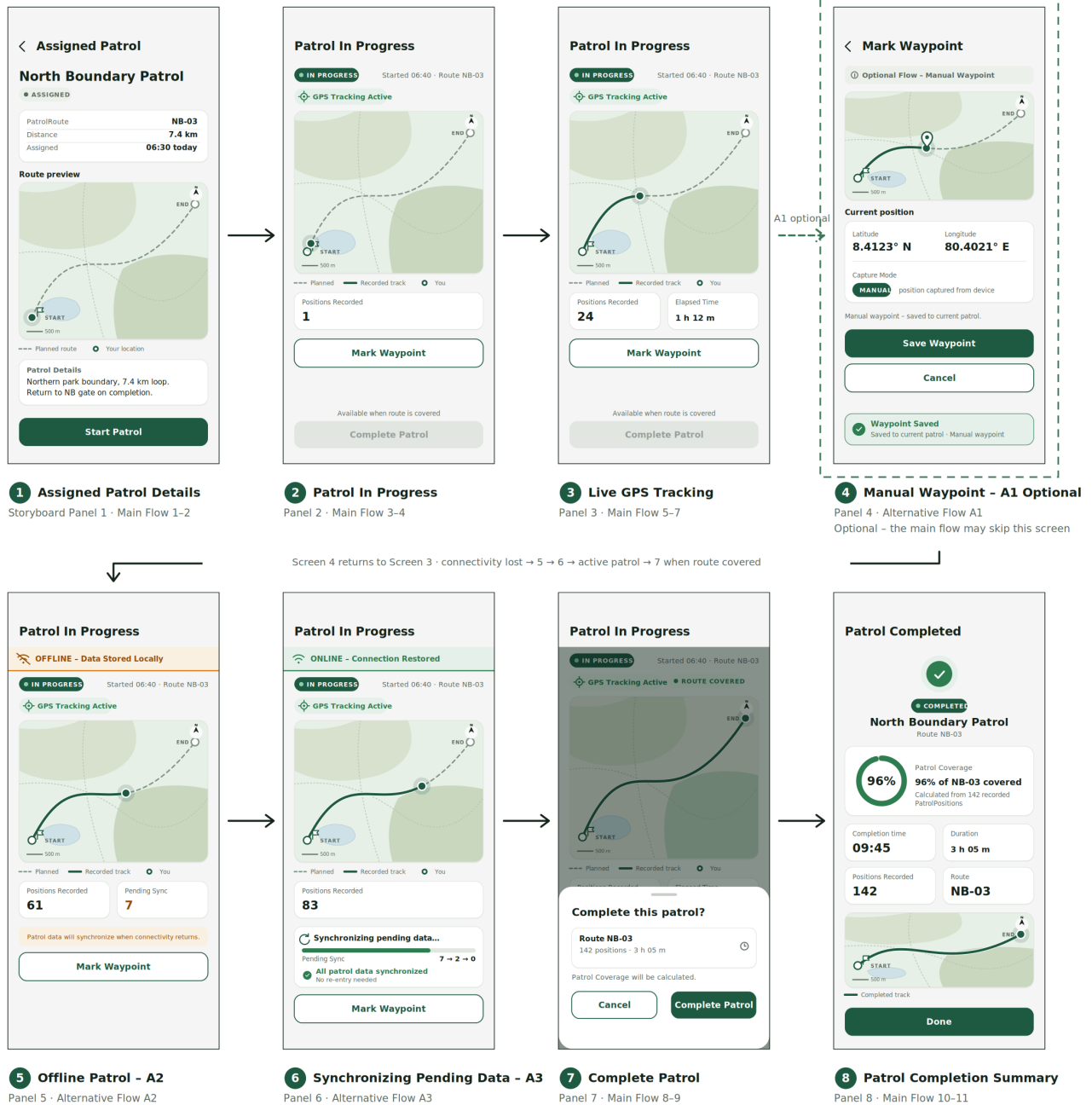
4.5 High-Fidelity Wireframe – UC01-S01

UC01-S01 - Conduct Assigned Ranger Patrol

High-Fidelity Wireframe – Ranger field application (mobile, 390 × 844)

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Visual system: primary #1F5A43 · secondary #3B7A57 · background #F6F8F5 · success #2E7D50 · offline amber #D97706. Every state has a text label + icon, never colour alone. Buttons 50-52 px; body text ≥ 13 px. Screen Structure, flow, terminology and functionality identical to SE3070_UC01-S01_LowFidelityWireframe_FINAL. Solid arrows = main success scenario; dashed = A1 (optional). Screens 5-6 = A2 / A3.

Figure 6 – High-fidelity wireframe for UC01-S01

5 Thushalini U (IT23794870) – UC02-S01 Report Field Incident

5.1 Use Case Scenario – UC02-S01 Report Field Incident

Use Case ID	UC02-S01
Use Case Name	Report Field Incident
Parent Use Case	UC02 – Report and Manage Field Incident
Goal / Description	This scenario describes how a Ranger reports a field incident encountered during a patrol using the field application. The Ranger selects the appropriate incident type, captures a photograph, records the current GPS location, and provides a short description of the incident. The System validates and records the incident information. If network connectivity is available, the incident and associated data are synchronised with the central system. If network connectivity is unavailable in the field, the System stores the incident data locally on the Ranger's device and marks it as pending synchronisation. When connectivity is restored, the System synchronises the pending incident data with the central system without requiring the Ranger to re-enter the information. This scenario supports the timely reporting of incidents such as snares, carcasses, illegal campsites, suspicious footprints, or other signs of illegal activity or wildlife risk.
Primary Actor	Ranger
Secondary Actor(s)	None
Trigger	The Ranger encounters a field incident during a patrol and decides to record and report it using the field application
Preconditions	P1. A valid Patrol has been assigned to the Ranger and is available on the field application. P2. The Ranger is authenticated and has access to the field application. P3. The Ranger has permission to create and submit field incident reports. P4. The field application is available on the Ranger's device. P5. The Ranger's device has access to camera and GPS/location capabilities, although individual GPS readings or photographs may occasionally be unavailable.

Postconditions – Success	S1. A FieldIncidentReport record is created and associated with the Ranger and the relevant Patrol. S2. The selected incident type is stored with the FieldIncidentReport. S3. The incident photograph is successfully attached to the FieldIncidentReport. S4. The captured GPS coordinates and time are stored with the FieldIncidentReport. S5. The incident description is stored with the FieldIncidentReport. S6. If the network is available, the FieldIncidentReport is synchronised with the central system. S7. If the network is unavailable, the FieldIncidentReport is preserved locally as pending synchronisation and is synchronised when connectivity is restored.
Postconditions – Minimal Guarantee	F1. Any FieldIncidentReport information successfully captured by the Ranger is preserved and is not lost, including information recorded while offline. F2. The FieldIncidentReport remains in a valid and consistent status even if submission or synchronisation temporarily fails

Main success scenario

Assumes normal network connectivity throughout. Alternative and exception flows below state the step from which they branch.

Step	Actor	Action
1	Ranger	Encounters a field incident during the assigned patrol and opens the Report Incident function in the field application.
2	System	Displays the Field Incident reporting screen with options to select an incident type, capture a photograph, record the location, and enter a description.
3	Ranger	Selects the appropriate incident type, such as Snare, Carcass, Illegal Campsite, Footprints, or another available incident category.
4	System	Records the selected incident type and associates it with the new FieldIncidentReport being prepared.
5	Ranger	Captures a photograph of the incident using the device camera.
6	System	Validates and attaches the captured photograph to the FieldIncidentReport.

Step	Actor	Action
7	System	Obtains the Ranger's current GPS position and records the latitude, longitude, and capture time.
8	Ranger	Enters a short description explaining the incident and any relevant observations.
9	System	Displays the entered incident type, photograph, GPS location, and description for review before submission.
10	Ranger	Reviews the incident information and selects Submit Incident.
11	System	Validates the required incident information and checks network connectivity.
12	System	Creates the FieldIncidentReport record and stores the incident information in the central system.
13	System	Associates the FieldIncidentReport with the Ranger and the active Patrol and stores the incident status as submitted.
14	System	Confirms successful submission to the Ranger and displays the incident reference/status.

Alternative flows

A1 – No Network Connectivity <i>Branches from step 11 of the main flow when the System detects that network connectivity is unavailable.</i>	A1.1 The System detects that network connectivity is unavailable. A1.2 The System displays an offline indicator to the Ranger and continues operating. A1.3 The Ranger confirms submission of the completed FieldIncidentReport. A1.4 The System creates the FieldIncidentReport information locally on the device. A1.5 The System stores the selected incident type, photograph, GPS coordinates, timestamp, description, Ranger information, and Patrol association locally. A1.6 The System marks the locally stored FieldIncidentReport as Pending Synchronisation. A1.7 The System displays the pending synchronisation status to the Ranger. A1.8 The Ranger continues the patrol without needing to re-enter the incident information.
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	A1.9 When network connectivity becomes available, Alternative Flow A2 – Connectivity Restored applies.
A2 – Connectivity Restored <i>Branches from Alternative Flow A1 when network connectivity becomes available again.</i>	A2.1 Network connectivity becomes available again. A2.2 The System identifies the locally stored FieldIncidentReport records that are pending synchronisation. A2.3 The System synchronises the pending FieldIncidentReport data with the central system. A2.4 The System uploads the associated photograph and incident information. A2.5 The System associates the synchronised FieldIncidentReport with the relevant Ranger and Patrol. A2.6 The System marks the successfully synchronised FieldIncidentReport as Synchronised. A2.7 The System reduces the pending synchronisation count accordingly. A2.8 The Ranger does not need to re-enter the previously recorded incident information. A2.9 The flow returns to the normal field operation.
A3 – Review and Edit Incident Before Submission <i>Branches from step 9 of the main flow when the Ranger identifies incorrect or incomplete information during review</i>	A3.1 The Ranger reviews the incident information and identifies an incorrect incident type, description, or photograph. A3.2 The Ranger selects the relevant field to edit. A3.3 The System allows the Ranger to modify the selected information. A3.4 The Ranger updates the information and returns to the review screen. A3.5 The System displays the updated incident information. A3.6 The flow returns to step 10 of the main flow.

Exception flows

E1 – GPS Position Cannot Be Obtained <i>Branches from step 7 of the main flow.</i>	E1.1 The System attempts to obtain the Ranger's current GPS position. E1.2 The System cannot obtain a valid GPS position, for example because of dense forest canopy, a ravine, or temporarily unavailable location services. E1.3 The System informs the Ranger that the current location could not be automatically recorded. E1.4 The Ranger may move to an area with better GPS availability and retry the location capture.
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	E1.5 If the Ranger's GPS position becomes available, the System records the coordinates and capture time. E1.6 If the location remains unavailable, the System preserves the other incident information and allows the Ranger to continue according to the application's incident reporting rules. E1.7 The FieldIncidentReport is not lost and the flow continues when the required location information can be successfully captured.
E2 – Photo Capture Fails <i>Branches from steps 5 of the main flow.</i>	E2.1 The Ranger attempts to capture a photograph of the incident. E2.2 The System cannot successfully capture or attach the photograph. E2.3 The System informs the Ranger that the photograph could not be recorded. E2.4 The Ranger may retry the photograph capture. E2.5 If the photograph is successfully captured, the System attaches it to the FieldIncidentReport. E2.6 The flow returns to step 7 of the main flow.
E3 – Required Information Missing <i>Branches from step 11 of the main flow.</i>	E3.1 The Ranger selects Submit Incident. E3.2 The System validates the FieldIncidentReport information. E3.3 The System identifies that one or more required fields are missing or invalid. E3.4 The System informs the Ranger which information must be completed. E3.5 The Ranger provides or corrects the required information. E3.6 The flow returns to step 9 of the main flow.
E4 – Incident Submission Fails <i>Branches from step 12 of the main flow when the central system cannot accept the incident.</i>	E4.1 The System attempts to create the FieldIncidentReport in the central system. E4.2 The submission fails because of a temporary server or communication problem. E4.3 The System informs the Ranger that the incident could not be submitted immediately. E4.4 The System preserves the complete FieldIncidentReport locally on the device. E4.5 The System marks the FieldIncidentReport as Pending Synchronisation. E4.6 The Ranger can continue the patrol without losing the recorded incident.



	E4.7 When connectivity and the central system become available, Alternative Flow A2 applies.
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5.2 Sequence Diagram – UC02-S01

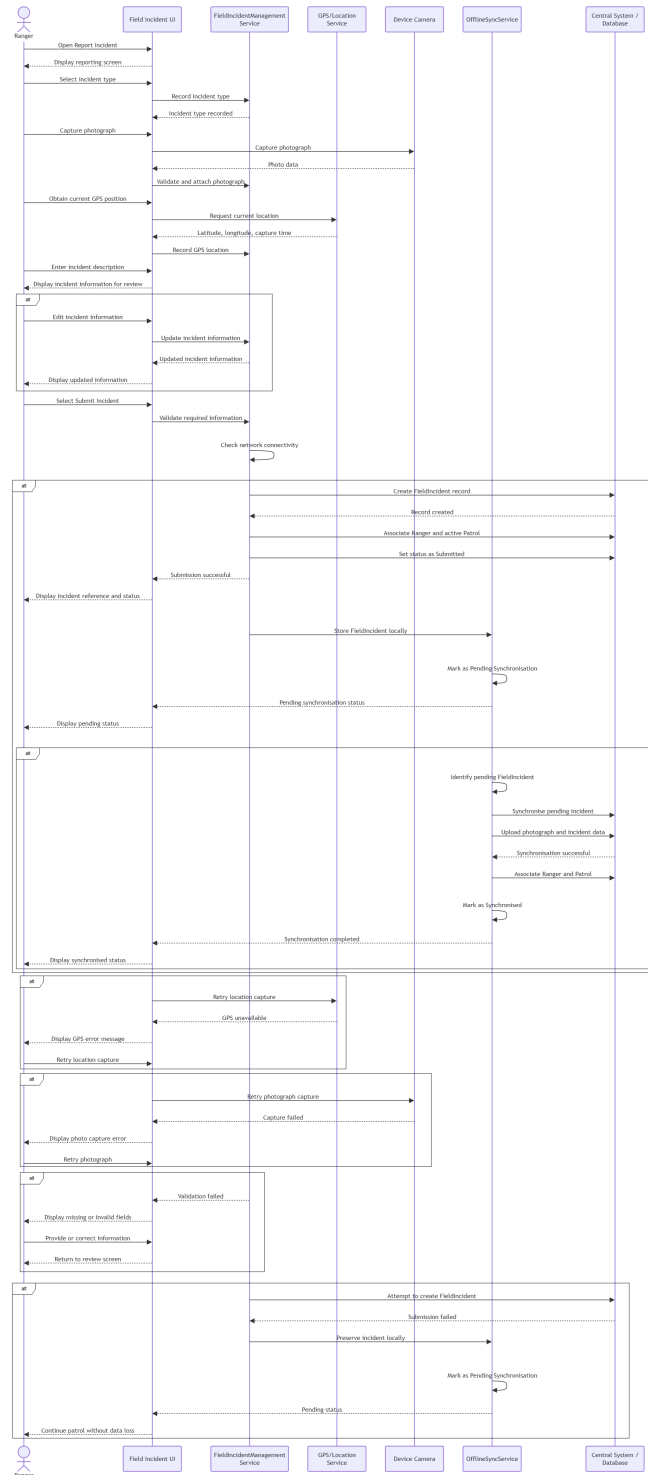


Figure 7 – Sequence diagram for UC02-S01 Report Field Incident

5.3 Hand-Drawn Storyboard – UC02-S01

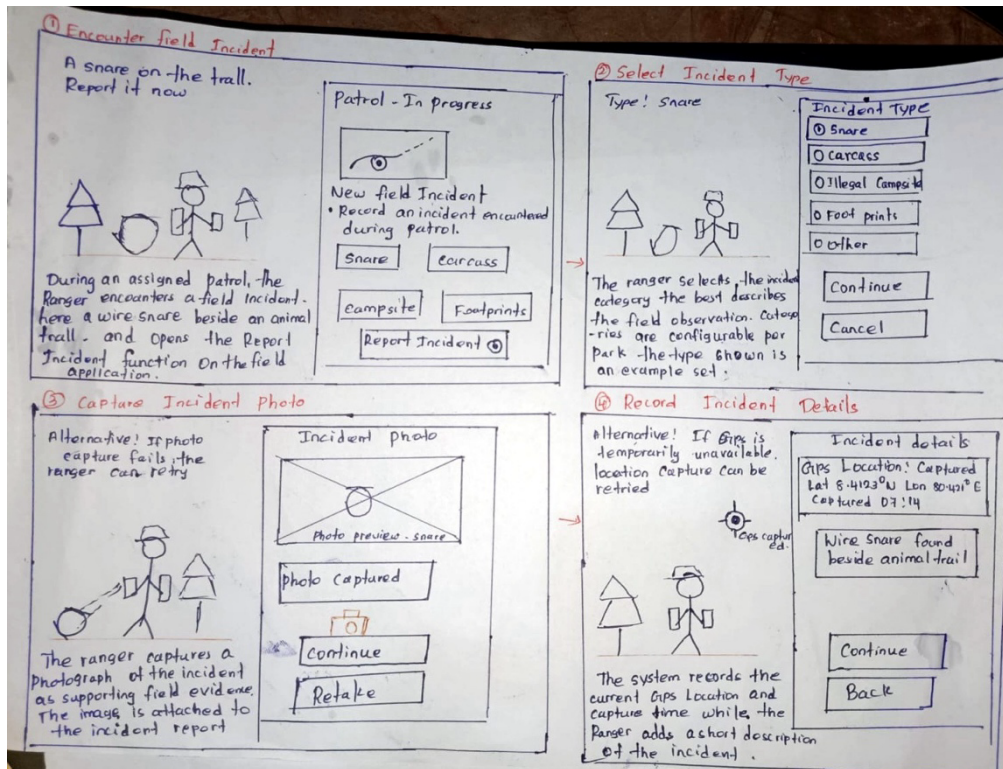


Figure 8(a) – Hand-drawn storyboard for UC02-S01 (panels 1–4)

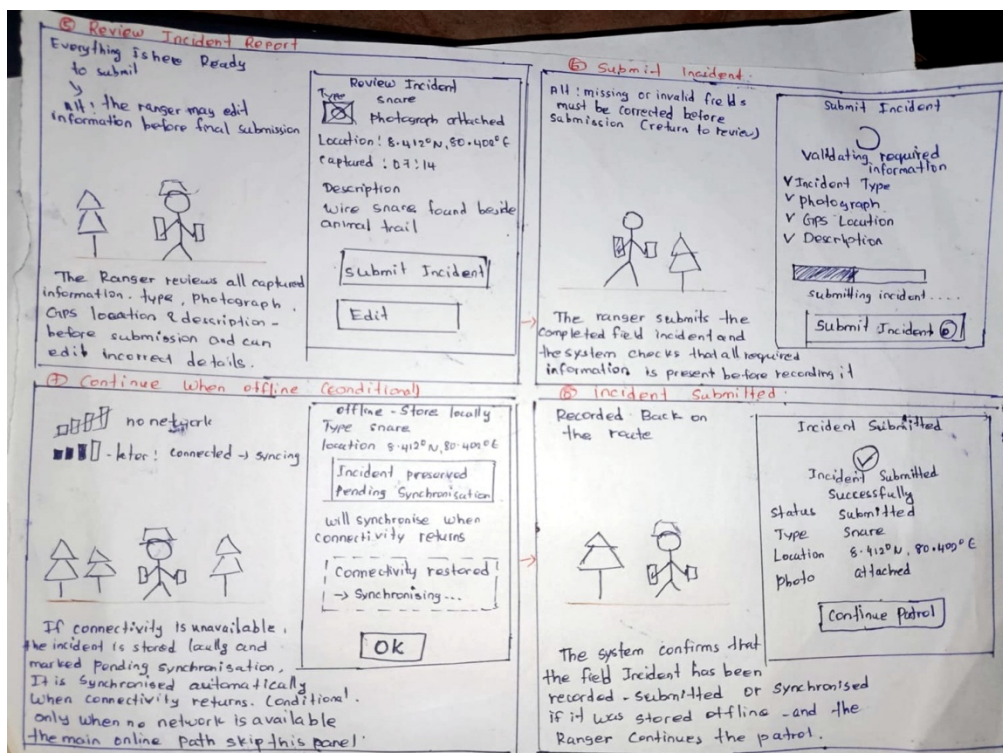


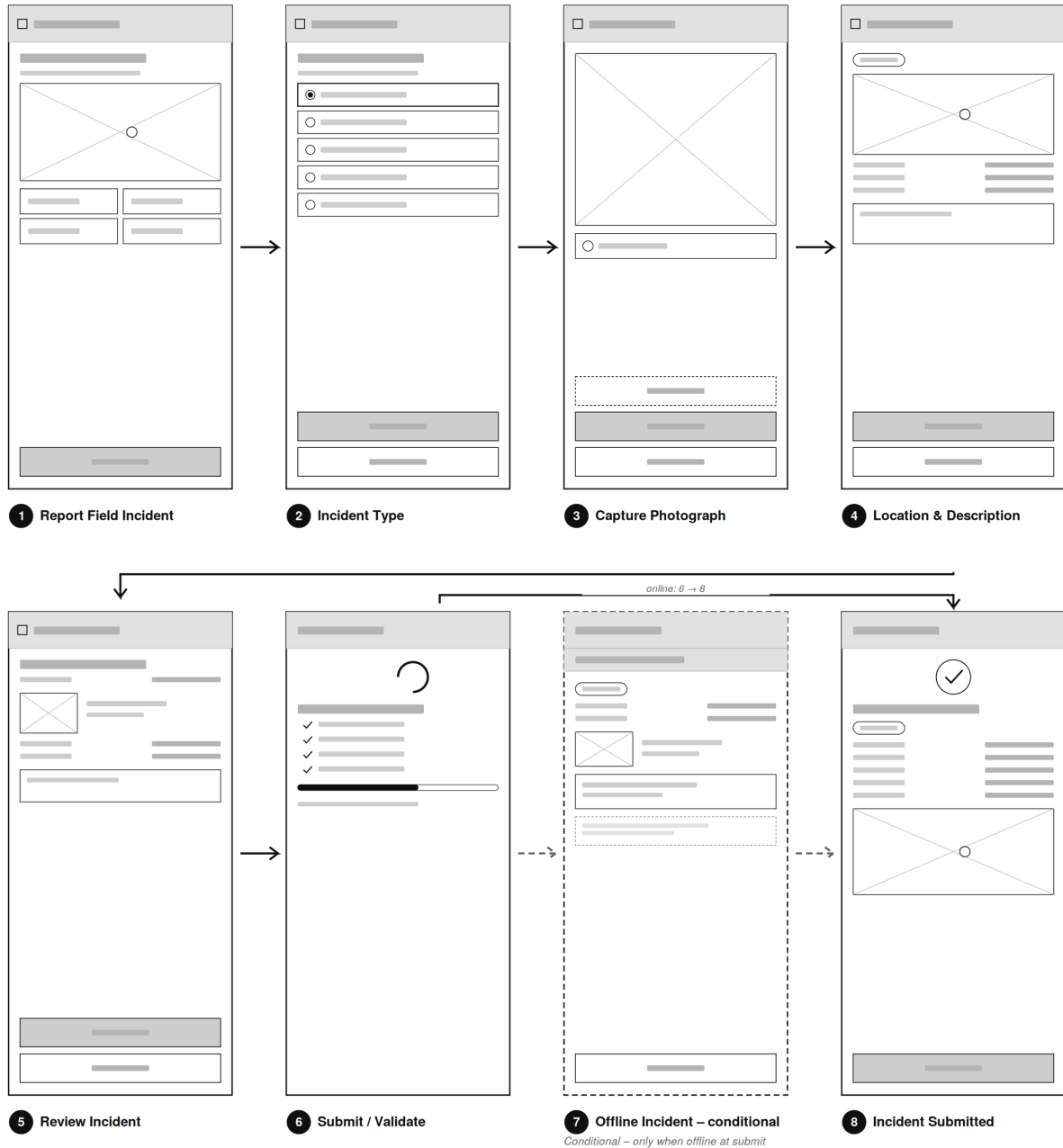
Figure 8(b) – Hand-drawn storyboard for UC02-S01 (panels 5–8)

5.4 Low-Fidelity Wireframe – UC02-S01

UC02-S01 – Report Field Incident

Low-Fidelity Wireframe – skeleton (screen rectangles 390 × 844, no device frame)

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Skeleton low-fidelity: layout, hierarchy, component placement and navigation only – grey bars stand for text, crossed boxes for Images / maps, filled rectangle = primary action, outlined = secondary, dashed = conditional. Screens are plain rectangles without a device frame. Content, labels and colour are defined in the high-fidelity wireframe of the same scenario.

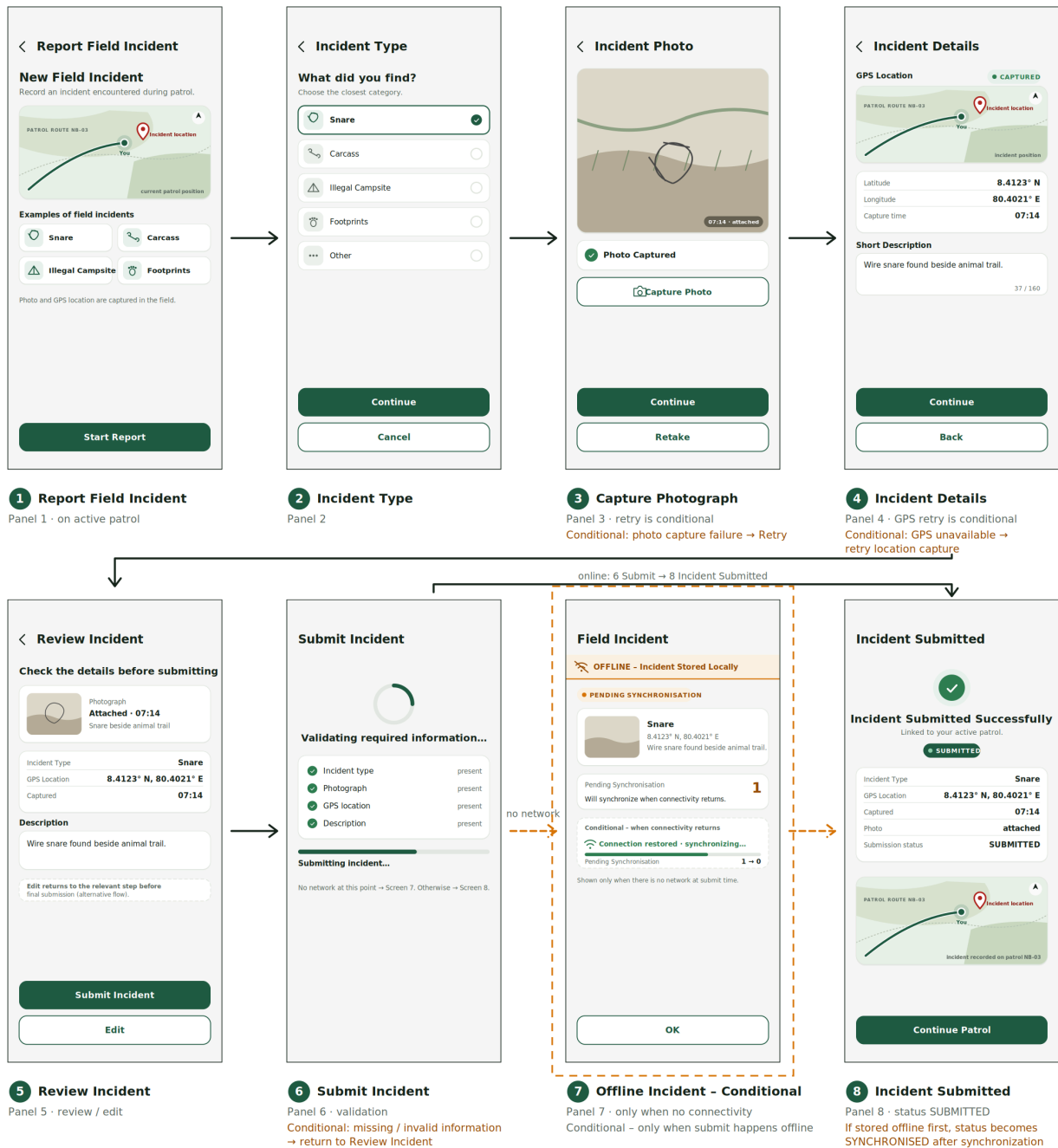
Figure 9 – Low-fidelity wireframe for UC02-S01

5.5 High-Fidelity Wireframe – UC02-S01

UC02-S01 - Report Field Incident

High-Fidelity Wireframe – Ranger field application (mobile, 390 × 844)

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Same design system as the group Ranger patrol UI; primary #1F5A43 · secondary #3B7A57 · background #F6F8F5 · success #2E7D50 · offline amber #D97706. Screens show the normal successful state; failures are noted in red. Structure and flow identical to Thushalini_UC02_FieldIncident_LowFidelityWireframe_FINAL. Dashed hint cards = conditional behaviour. Every state has label + icon, never colour alone. Buttons 50–52 px. Screens shown as per design.

Figure 10 – High-fidelity wireframe for UC02-S01

6 Vaishnavi L (IT23717336) – UC03-S01 Monitor Tracked Wildlife and Manage Risk Alerts

6.1 Use Case Scenario – UC03-S01 Monitor Tracked Wildlife and Manage Risk Alerts

Use Case ID	UC03-S01
Use Case Name	Monitor Tracked Wildlife and Manage Risk Alerts
Primary Actor(s)	Ranger, Community Liaison Officer
Secondary Actor(s)	GPS Tracking Collar (sensor), Operations Dashboard (park manager view)
Trigger	A GPS tracking collar transmits a periodic location update for a monitored animal (typically an elephant).
Description	This use case describes how the system continuously receives location data from animal-borne GPS collars, evaluates whether a tracked animal has entered a high-risk zone such as farmland or a road near a village boundary, and — if so — automatically generates and dispatches a risk alert to the nearest ranger and the community liaison officer so that human-elephant conflict can be prevented before it occurs.
Preconditions	PRE-1: The animal is fitted with an active, registered GPS collar that is transmitting. PRE-2: The park's high-risk zone boundaries (farmland, roads, village edges) have been configured in the system. PRE-3: At least one ranger and one community liaison officer are registered and available to receive notifications for the relevant park sector.
Postconditions (Success)	POST-1: A risk alert record is created, time-stamped, and linked to the animal, zone, and location. POST-2: The ranger and community liaison officer have been notified and the alert status is tracked (New → Acknowledged → Resolved). POST-3: The operations dashboard and historical conflict records are updated for reporting.
Postconditions (Failure)	No alert is raised when no risk is detected; the location update is simply logged for patrol/movement history.

Main flow (basic path)

Step	Action
1	The GPS collar periodically transmits the animal's current location (collar ID, latitude, longitude, timestamp) to the Monitoring Service.
2	The Monitoring Service forwards the location to the Risk Zone Checker for evaluation.
3	The Risk Zone Checker compares the animal's position against the configured high-risk zone boundaries for that park.
4	The Risk Zone Checker determines that the animal has entered a high-risk zone (e.g., farmland or a road) and reports this back to the Monitoring Service.
5	The Monitoring Service requests the Alert Manager to generate a new risk alert, including the animal ID, zone type, severity, and location.
6	The Alert Manager creates an alert record and pushes it to the Operations Dashboard.
7	The Alert Manager sends a push notification to the nearest on-duty ranger and to the community liaison officer responsible for that boundary sector.
8	The ranger opens the alert on the mobile app, reviews the map, animal details, and distance, then acknowledges the alert.
9	The alert status is updated to "Acknowledged" on the operations dashboard.
10	The ranger (optionally together with the community liaison officer) travels to the location and takes action to move the animal away or warn nearby residents.
11	The ranger records the outcome and marks the alert as "Resolved" from the mobile app.
12	The alert status and outcome notes are synchronized to the operations dashboard for reporting and pattern analysis.

Alternative flows

AF-1: Animal remains in a safe zone (Step 4)	The Risk Zone Checker determines the animal is not within a high-risk zone. No alert is generated; the location is written to the animal's movement history for later analysis. The use case ends.
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AF-2: Ranger does not respond within the configured time window (Step 8)	If the alert is not acknowledged within a configured threshold (e.g., 10 minutes), the Alert Manager escalates the alert to the next-nearest ranger and/or the park manager. The escalation is logged against the alert record. Flow resumes at Step 8 with the newly notified ranger.
AF-3: Camera trap corroboration	If a nearby motion-triggered camera trap has recently uploaded an image of the same area, the system attaches the most recent image to the alert for visual confirmation before the ranger sets out.

Exception flows

EF-1: Collar signal lost or delayed	If no location update is received from a registered collar within the expected interval, the system flags the collar as “Signal Lost” and notifies the park manager rather than raising a risk alert.
EF-2: No network connectivity on ranger's device	If the ranger's mobile device has no connectivity when the alert is pushed, the notification is queued and delivered as soon as the device reconnects; meanwhile the community liaison officer's notification (if connected) still proceeds so a response is not entirely blocked.
EF-3: Duplicate or conflicting alerts	If a new location update arrives for an animal that already has an open, unresolved alert, the system updates the existing alert's location trail instead of creating a duplicate alert record.

Business rules / special requirements

- High-risk zone boundaries must be independently configurable per park, since Sinharaja's dense jungle and Yala's open grassland require different zone shapes and alert sensitivity.
- Alert notifications must be delivered as push notifications where data connectivity exists, and must not depend on a live connection for the ranger to view previously synced alert data.
- All alert and location data generated in the field must be cached locally on the mobile device and synchronized automatically once connectivity is restored.
- Every alert must retain a full audit trail of status changes (New, Acknowledged, Resolved, Escalated) with timestamps and the responsible user, to support later reporting to funding bodies and government ministries.

6.2 Sequence Diagram – UC03-S01

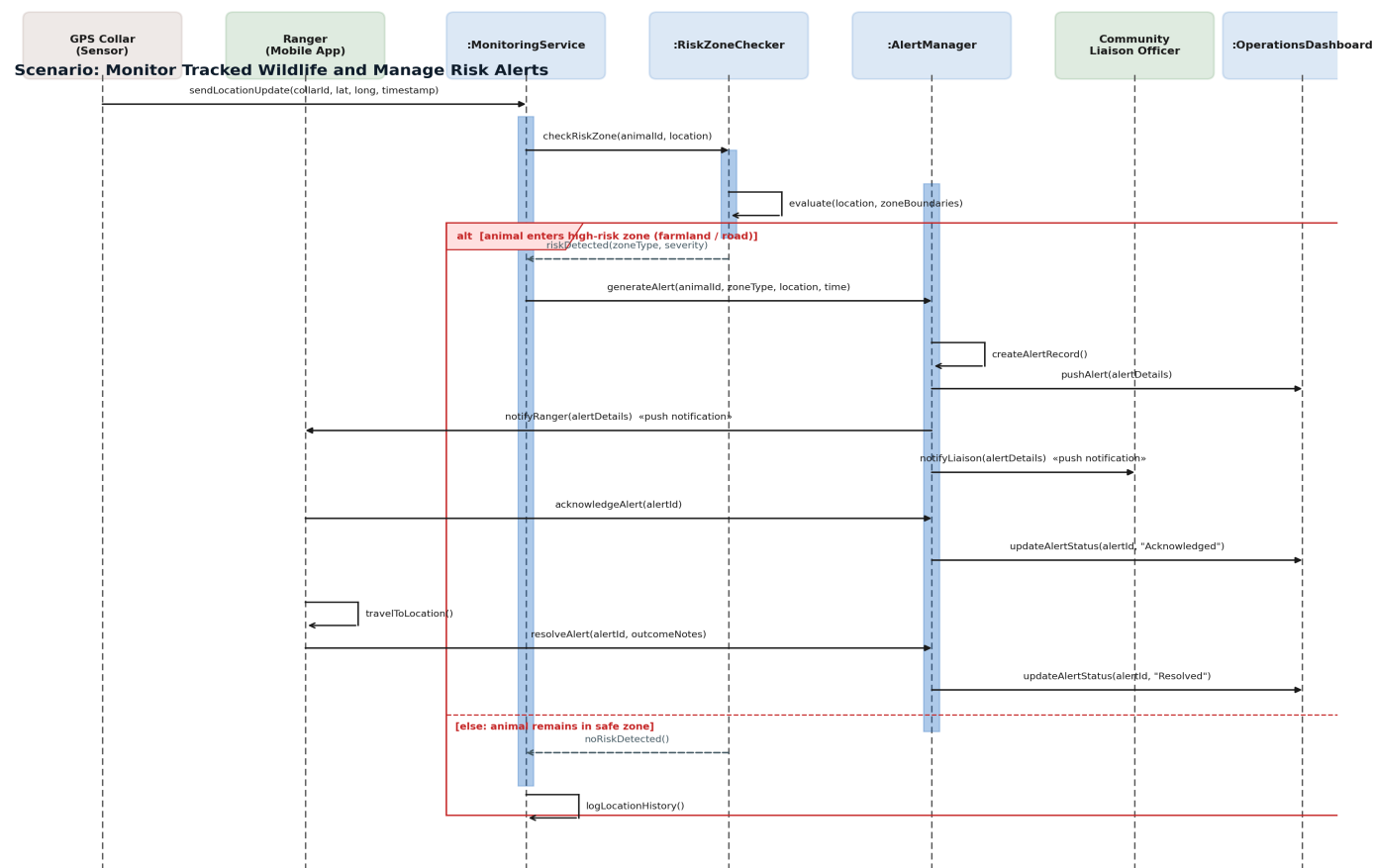


Fig. 1 — Sequence Diagram: Monitor Tracked Wildlife and Manage Risk Alerts

Figure 11 – Sequence diagram for UC03-S01 Monitor Tracked Wildlife and Manage Risk Alerts

6.3 Hand-Drawn Storyboard – UC03-S01

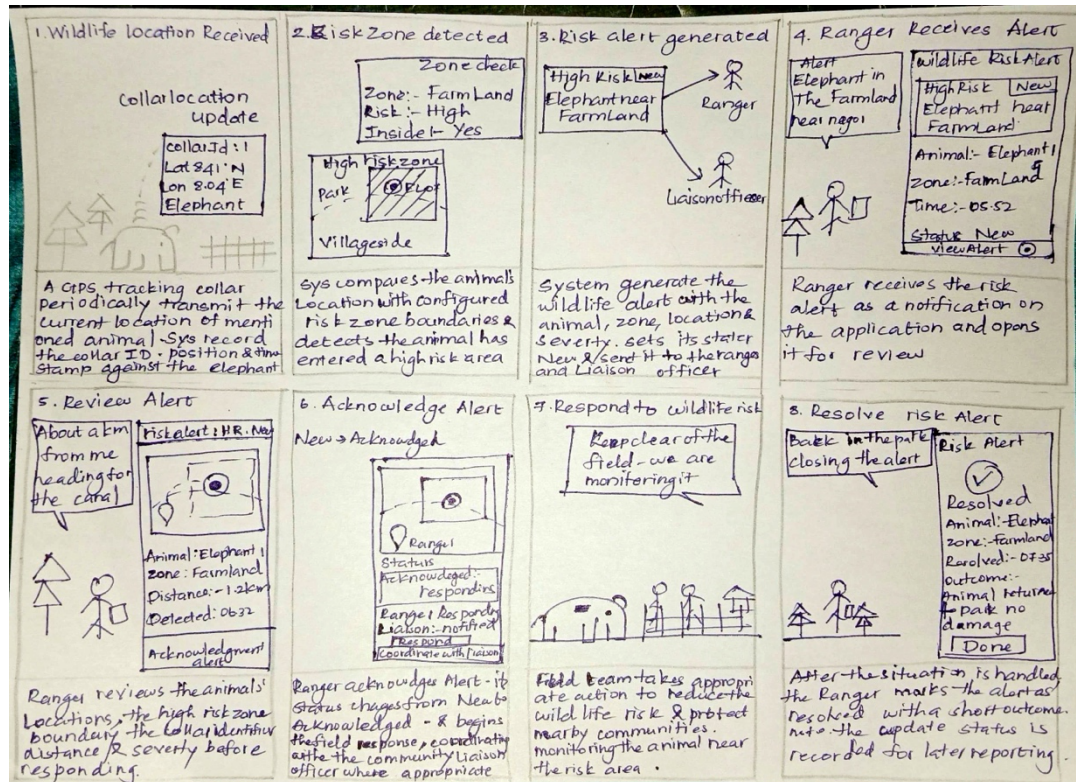


Figure 12 – Hand-drawn storyboard for UC03-S01

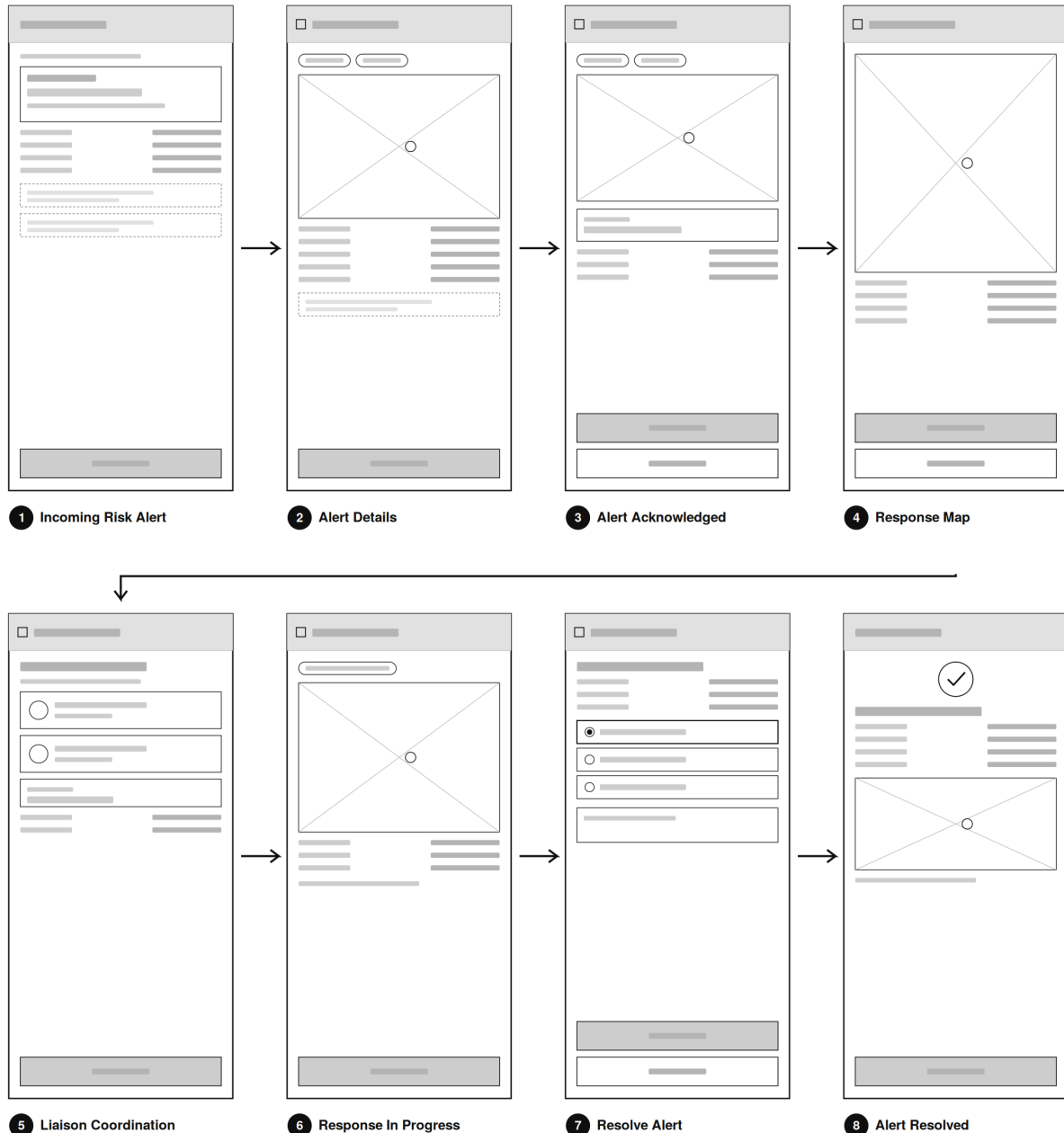
6.4 Low-Fidelity Wireframe – UC03-S01

UC03-S01 – Monitor Tracked Wildlife and Manage Risk Alerts

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Low-Fidelity Wireframe – skeleton (screen rectangles 390 × 844, no device frame)

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Skeleton low-fidelity: layout, hierarchy, component placement and navigation only – grey bars stand for text, crossed boxes for images / maps, filled rectangle = primary action, outlined = secondary, dashed = conditional. Screens are plain rectangles without a device frame. Content, labels and colour are defined in the high-fidelity wireframe of the same scenario.

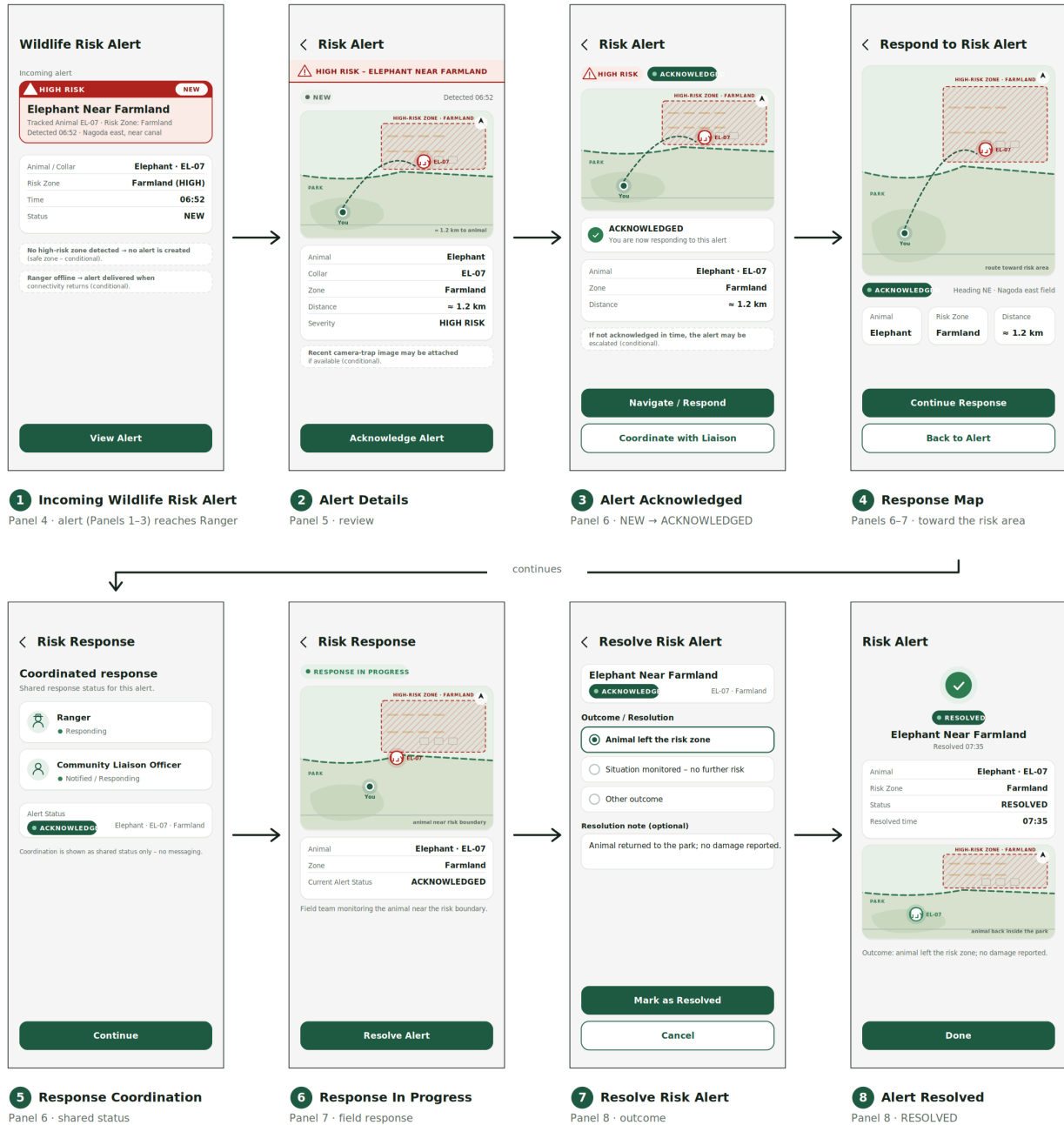
Figure 13 – Low-fidelity wireframe for UC03-S01

6.5 High-Fidelity Wireframe – UC03-S01

UC-03 - Monitor Tracked Wildlife and Manage Risk Alerts

High-Fidelity Wireframe – Ranger field application (mobile, 390 × 844)

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Same design system as the group Ranger patrol UI: primary #1F5A43 · secondary #3B7A57 · background #F6F8F5 · success #2E7D50 · red used only for the HIGH RISK alert state and risk-zone overlay. Structure and flow identical to Vaishnavi_UC03_WildlifeRiskAlerts_LowFidelityWireframe_FINAL. Dashed hint cards = conditional behaviour. Every state has label + icon, never colour alone. Buttons 50-52 px. Screens shown

Figure 14 – High-fidelity wireframe for UC03-S01

7 Kanistan T (IT23748644) – UC04-S01 Manage Human-Wildlife Conflict Reports

7.1 Use Case Scenario – UC04-S01 Manage Human-Wildlife Conflict Reports

Use Case ID	UC04-S01
Use Case Name	Manage Human-Wildlife Conflict Reports
Description	This use case scenario describes the process by which a community member living near a park boundary reports a human-wildlife conflict, such as an elephant sighting or crop-raiding incident, using a mobile application or SMS reporting channel. The report is received by the wildlife conservation system and made available to rangers or community liaison officers through the central operations dashboard for response and status tracking.
Preconditions	The wildlife conservation reporting system is available. The community member is located near a park boundary. The community member has access to the mobile reporting application or SMS service. The reporting channel is configured to receive community reports. The operations dashboard is available for authorized staff. Rangers or community liaison officers are available to review and respond to reports.
Postconditions	The human-wildlife conflict report is stored in the system. The report is available through the central operations dashboard. A ranger or community liaison officer can review the report. The response to the incident is recorded. The status of the report is updated. The report becomes part of the historical data used to identify human-wildlife conflict trends.

Main flow

Step	Action
1	Community member opens the wildlife reporting mobile application or uses the SMS reporting channel.
2	System displays/prompts for the required incident information.

Step	Action
3	Community member selects the incident type, such as Elephant Sighting or Crop Raiding.
4	Community member provides the location of the incident.
5	Community member enters a short description of the incident.
6	Community member submits the report.
7	System validates the submitted information.
8	System creates and stores the human-wildlife conflict report.
9	System sends the report to the central operations dashboard.
10	Ranger or community liaison officer reviews the reported incident.
11	Ranger or community liaison officer determines and initiates the appropriate response.
12	Ranger or community liaison officer updates the report status.
13	System stores the updated status and response information.

Alternate flows

Report Submitted Through Mobile Application	Community member uses the mobile application to enter the incident details. System receives and processes the report. Report is displayed on the central operations dashboard.
Report Submitted Through SMS	Community member sends the incident information through the SMS short code. System receives and processes the SMS report. System creates a conflict report from the received information. Report is displayed on the central operations dashboard.
Multiple Reports	Several community members report incidents from the same or nearby area. System stores each report separately. Operations staff can review the reports and identify possible conflict patterns.

Exception flows



No Mobile Network / Connectivity	Community member attempts to submit a report while connectivity is unavailable. System stores the report locally where offline support is available. Once connectivity becomes available, the report is synchronized with the central system. The report is then made available on the operations dashboard.
Incomplete Report	Community member submits a report without required information. System identifies the missing information. System prompts the community member to provide the required details. Report is submitted after the required information is provided.
System Error	System encounters an error while processing the report. System displays an error message. Report is not marked as successfully submitted until it can be processed.
Dashboard Unavailable	A valid report is received but the operations dashboard is temporarily unavailable. System stores the report. Report becomes available to operations staff once the dashboard/system connection is restored.

7.2 Sequence Diagram – UC04-S01

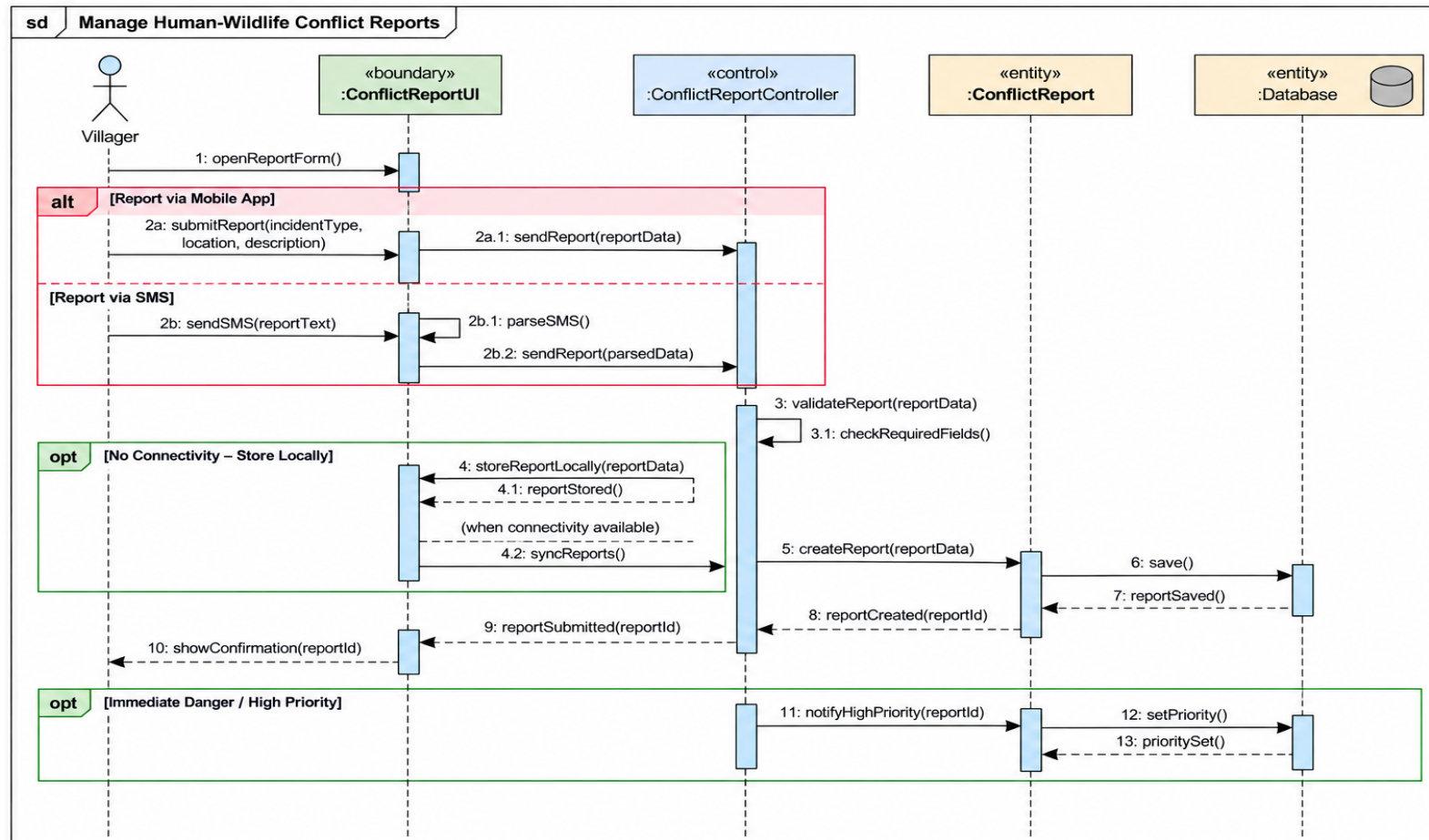


Figure 15 – Sequence diagram for UC04-S01 Manage Human-Wildlife Conflict Reports

7.3 Hand-Drawn Storyboard – UC04-S01

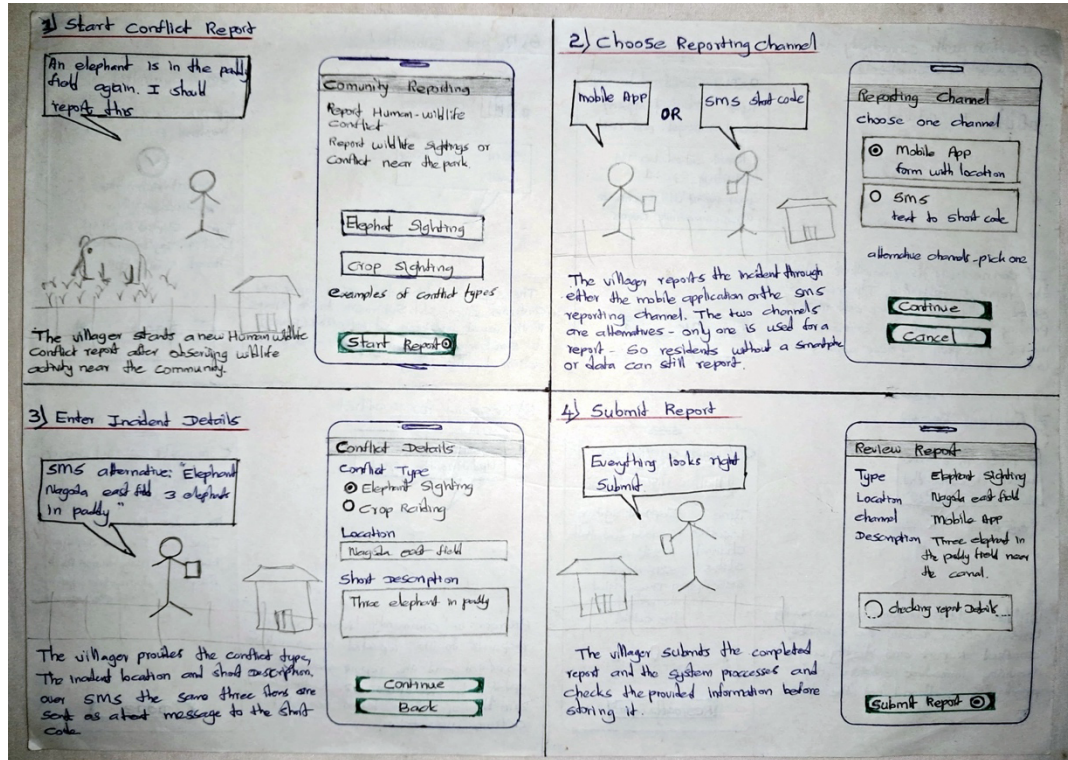


Figure 16(a) – Hand-drawn storyboard for UC04-S01 (panels 1–4)

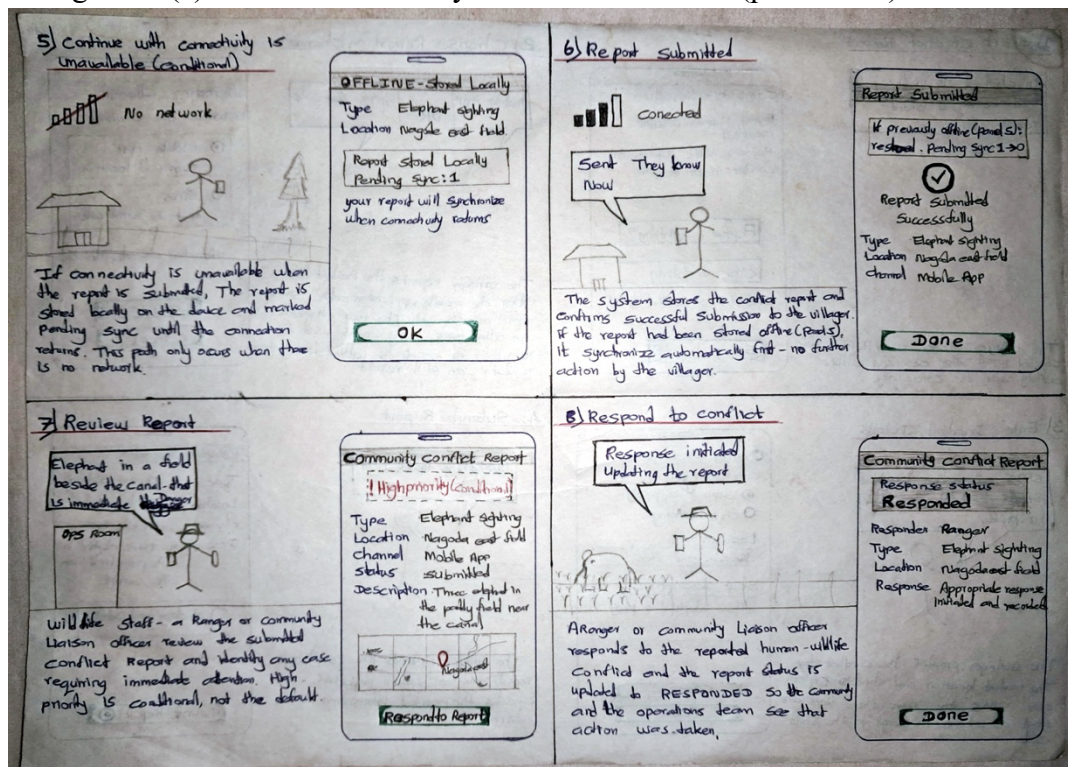


Figure 16(b) – Hand-drawn storyboard for UC04-S01 (panels 5–8)

7.4 Low-Fidelity Wireframe – UC04-S01

UC04-S01 – Manage Human-Wildlife Conflict Reports

Low-Fidelity Wireframe – skeleton (screen rectangles 390 × 844, no device frame)

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Skeleton low-fidelity: layout, hierarchy, component placement and navigation only – grey bars stand for text, crossed boxes for images / maps, filled rectangle = primary action, outlined = secondary, dashed = conditional. Screens are plain rectangles without a device frame. Content, labels and colour are defined in the high-fidelity wireframe of the same scenario.

Figure 17 – Low-fidelity wireframe for UC04-S01

7.5 High-Fidelity Wireframe – UC04-S01

Manage Human-Wildlife Conflict Reports

High-Fidelity Wireframe - community reporting and staff response (mobile, 390 × 844)

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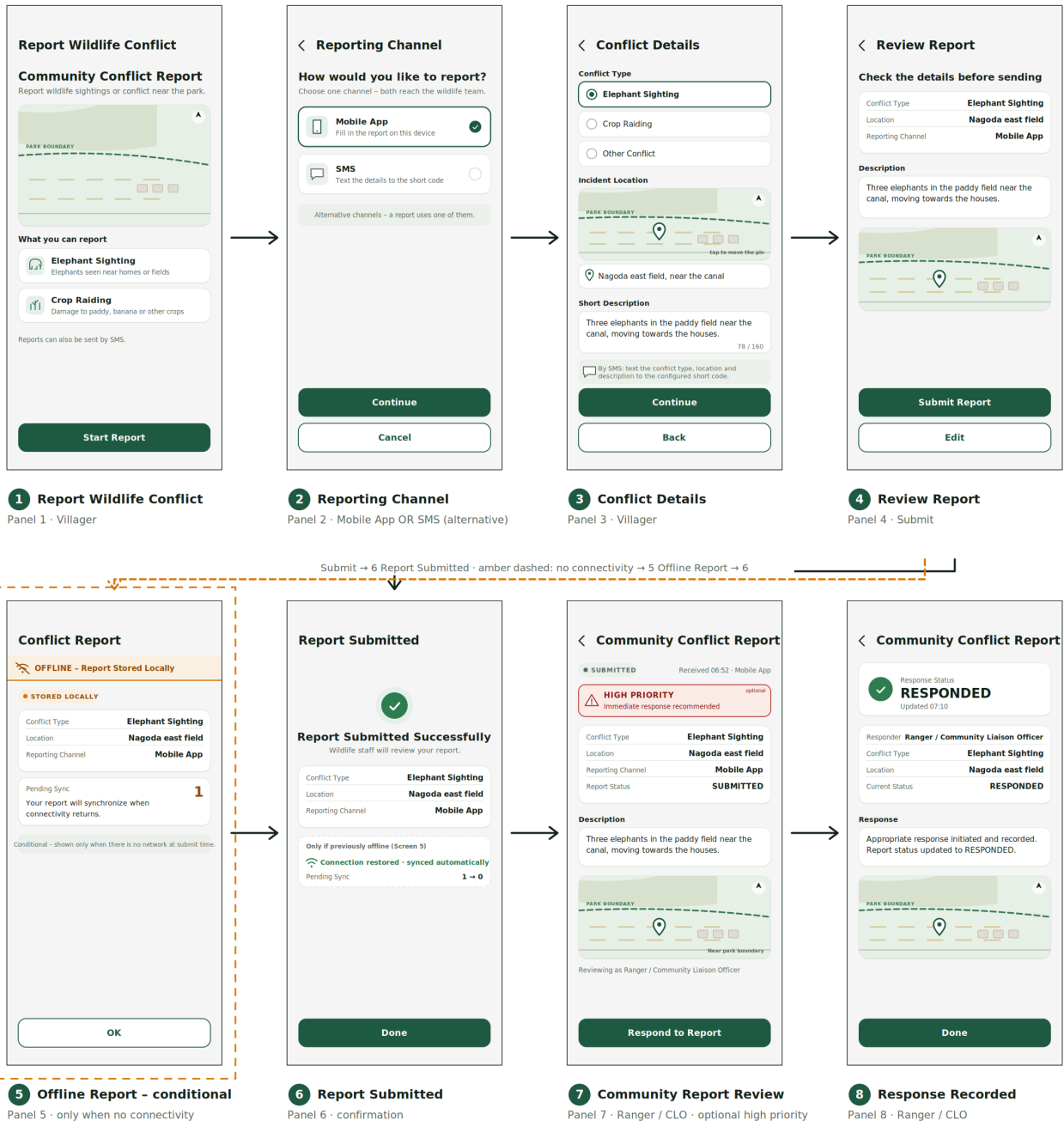


Figure 18 – High-fidelity wireframe for UC04-S01